

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Winter 2022 - 23 Examination

Semester : 5
Subject Code: 03105303
Subject Name: Theory of Computation

Date: 14/10/2022
Time: 10:30am to 1:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (Fill in the blanks, one word answer, MCQ-not more than Five (15) in case of MCQ) (All are compulsory) (Each of one mark)

1. Push down machine represents
 - a) Type-0 Grammar
 - b) Type-1 Grammar
 - c) Type-2 Grammar
 - d) Type-3 Grammar
2. The behavior of a NFA can be simulated by DFA
 1. Always
 2. Sometimes
 3. Never
 4. Depends on NFA
3. The logic of pumping lemma is a good example of
 - a) The pigeon hole principle
 - b) Divide and conquer method
 - c) Iteration
 - d) Recursion
4. If every string of a language can be determined whether it is legal or illegal in finite time the language is called
 - a) Decidable
 - b) Undecidable
 - c) Interpretive
 - d) Non Deterministic
5. Context free language can be recognized by
 - a) Finite state Automaton
 - b) Linear bounded automaton
 - c) Pushdown Automaton
 - d) Both b and c
6. Context-free languages are closed under Union True/False
7. The language accepted by finite automata is ----- (Regular language, Context free language, Context sensitive language)
8. Finite state machine _____ recognize palindromes. (Can, Cannot, May, May not)
9. If two finite state machines are equivalent they should have the same number of _____
10. What is Pumping Lemma ?
11. What is the difference between Moore and Mealey machines?
12. What is intersection of two FA's ?
13. What are Turing Machines ?
14. Discuss the difference between Context free languages and regular languages?
15. Define a recursively enumerable language

Q.2 Answer the following questions. (Attempt any three)

(15)

A) Differentiate between NFA , DFA using tuple representation

B) Prove that $\sqrt{2}$ is Irrational by method of Contradiction

C) Draw a FA for following regular language.

(i) $(11+110)^* 0$

(ii) $(0+1)^*(10+11)$

D) Prove that the following language is ambiguous

$S \rightarrow S + S \mid S * S \mid a$

Q.3 A) Write Regular Expressions for the following languages of all strings in $\{0,1\}^*$

(07)

(i) Strings that do not end with 01.

(ii) The language of all strings containing both 101 and 010 as substrings

B) What is Context Free Grammar(CFG). Define a PDA for CFG and give the tuple representation for the same.

(08)

OR

B) Given the Context Free Grammar G, find a CFG G' in Chomsky Normal Form generating $L(G) - \{ \}$

(08)

$S \rightarrow ABa$

$A \rightarrow aab$

$B \rightarrow Ac$

Q.4 A Design Turing machine to accept language $L = \{ WW^R \mid W \in (a+b)^* \}$.

(07)

OR

A) Define Turing machine. Explain with diagram, general structure of multi tape Turing machine. [6]

(07)

B) Write PDA for the string with equal number of a's and b's. Trace it with an example.

(08)